LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 3 Test (Set B): Metals and Non-metals

Chapter / Topic: Metals and Non-metals

For teacher use only. Do not distribute to students.

SECTION A

Q1. Which non-metal is a liquid at room temperature? **[1]**

Answer: (c) Bromine (mercury is a liquid metal)

Q2. The property of metals that allows them to be drawn into thin wires is: **[1]**

Answer: (d) ductility

Q3. Which of these metals reacts only with steam, not with cold or hot water? **[1]**

Answer: (b) Iron

Q4. In electrolytic refining, the anode mud consists of: **[1]**

Answer: (b) insoluble impurities

Q5. Which of the following oxides is basic? **[1]**

Answer: (d) Na_2O

Q6. **Assertion (A):** Ionic compounds are generally solids and somewhat hard. **Reason (R):** There is a strong force of attraction between the positive and… **[1]**

Answer: (a)

SECTION B

Q7. Why is sodium kept immersed in kerosene oil? **[2]**

Answer: Sodium reacts so vigorously with oxygen and moisture in the air that it catches fire if kept in the open. Kerosene keeps air and moisture away from it, preventing accidental fires.

- Very reactive with air/moisture, catches fire: 1
- Kerosene keeps air and water away: 1

OR, answer: Anodising is a process of forming a thick oxide layer on aluminium. The aluminium article is made the anode and electrolysed with dilute sulphuric acid. The thicker oxide layer protects it from further corrosion and can also be dyed to give an attractive finish.

- Process: 1
- Purpose: 1

Q8. Name two metals that are the best conductors of heat, and two that are comparatively poor conductors of heat. **[2]**

Answer: Best conductors: silver and copper. Comparatively poor conductors: lead and mercury.

- Each pair: 1

SECTION C

Q9. Element X has atomic number 12, and element Y has atomic number 17. **[3]**

(i) X: 2, 8, 2 (magnesium). Y: 2, 8, 7 (chlorine). [1]

(ii) XY_2 ($MgCl_2$). X loses 2 electrons and each of two Y atoms gains 1 electron. [1]

(iii) Ionic bond. Properties (any two): high melting point; soluble in water; conducts electricity when molten or in solution. [1]

Q10. Describe an activity to compare the reactivity of magnesium, aluminium, zinc and iron with dilute hydrochloric acid. Write the equations for… **[3]**

Answer: Take four test tubes with the same volume of dilute HCl and add a clean sample of each metal. Observe how fast bubbles of hydrogen form, and how warm each tube gets. Magnesium reacts most vigorously, then aluminium, zinc and iron. $Mg + 2HCl \rightarrow MgCl_2 + H_2$ and $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$

- Activity and observation: 1
- Order of reactivity: 1
- Equations: 1

OR, answer: Mineral: an element or compound that occurs naturally in the Earth's crust. Ore: a mineral from which a metal can be extracted profitably. Gangue: the unwanted impurities, such as soil and sand, present in an ore.

- Each: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Copper is low in the reactivity series, so heating alone works. The ore is heated in air, and part of it changes into copper(I) oxide: $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$. The copper(I) oxide then reacts with the remaining copper sulphide to give copper metal: $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$ [3]

- Heating alone (low reactivity): 1
- Each equation: 1 + 1

(ii) These metals have a greater affinity for oxygen than carbon has, so carbon cannot remove oxygen from their oxides. They are obtained by electrolysis of their molten chlorides (electrolytic reduction). [2]

- Reason: 1
- Electrolysis: 1

OR, answer: (a) (i) It is easier to obtain a metal from its oxide than from its carbonate or sulphide. (ii) Aluminium is light, a good conductor of heat, and protected from corrosion by a thin oxide layer. (iii) They are lustrous, malleable, and do not corrode or react with air and water. (b) Solder is an alloy of lead and tin. It has a low melting point, so it melts easily to join wires.

- (a) 1 each
- (b) composition 1, low melting point 1

SECTION E

Q12. When iron(III) oxide is mixed with aluminium powder and ignited, the reaction gives out so much heat that the iron formed is molten. This molten iron… **[4]**

(i) $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + \text{heat}$ [1]

(ii) Aluminium (it takes oxygen away from iron(III) oxide). [1]

(iii) Aluminium is more reactive than iron, so it can displace iron from its oxide. It is a displacement reaction and a redox reaction (and it is highly exothermic). [2]

- Al more reactive: 1
- Displacement + redox: 1